

# **T660**

## **DIGITAL DELAY GENERATOR**



## **Technical Manual**

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## **1 Introduction**

The T660 is a small, enclosed digital delay/pulse generator which is intended for use in embedded OEM applications. Given an internal or external trigger, it outputs four precisely timed pulses. It is a drop-in, backward compatible replacement for the T560 and T564 digital delay and pulse generators.

Standard features include:

- Four TTL-level delay outputs, individually programmable for delay and pulse width
- 10 picosecond delay and width resolution, 60 second range
- 21 nanosecond  $\pm$  400 picosecond insertion delay, 16 MHz max trigger rate
- <35 picoseconds typical jitter
- Internal 10 MHz crystal oscillator time base with external lock capability
- 0-16 MHz DDS synthesizer for internal trigger rates
- Programmable-level trigger input with divide and burst features and trigger enable GATE input
- RS-232 serial and 10/100 Mbps Ethernet interfaces
- Extruded enclosure with removable mounting flange
- Output impedance switch allowing for 50  $\Omega$  source termination or low impedance
- Trains/Frames advanced option (T660-2) includes timing lists and multiple pulses per trigger
- Firmware upgradable from any browser
- 12V power supply furnished with purchase

## 2 Specifications

|                 |                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FUNCTION        | Four channel digital delay/pulse generator                                                                                                                                                                                                                                                                                              |
| CHANNELS        | A B C D                                                                                                                                                                                                                                                                                                                                 |
| OUTPUT VOLTAGE  | +5V<br>Zout selectable: <ul style="list-style-type: none"> <li>• 50 <math>\Omega</math> (+2.5V into 50 <math>\Omega</math> loads)</li> <li>• Rise/Fall &lt; 1.5 ns typ.</li> <li>• low z (&gt;4V into 50 <math>\Omega</math> loads)</li> <li>• Rise/Fall &lt; 750 ps typ.</li> </ul>                                                    |
| DELAY RANGE     | 0 to 3600 seconds                                                                                                                                                                                                                                                                                                                       |
| WIDTH RANGE     | 0 to 3600 seconds                                                                                                                                                                                                                                                                                                                       |
| INSERTION DELAY | Trigger to any output 21 ns $\pm$ 400 ps                                                                                                                                                                                                                                                                                                |
| REP RATE        | 0 to 16 MHz, limited to 1/ (max d + w + 62.5 ns)                                                                                                                                                                                                                                                                                        |
| ACCURACY        | Trigger to rising or falling edges $\pm$ 400 ps $\pm$ 7.5 ps/ $^{\circ}$ C $\pm$ clock accuracy                                                                                                                                                                                                                                         |
| RESOLUTION      | Edge times, 10 ps<br>Trigger level, 10 mV                                                                                                                                                                                                                                                                                               |
| JITTER          | Typical 35 ps RMS + timebase jitter<br>Max 50 ps RMS + timebase jitter<br>Add clock jitter for delays > 500 $\mu$ s                                                                                                                                                                                                                     |
| TRIGGER         | External, internal, software (remote command)<br>Burst, divide-by-N, N-of-M pulse picking<br>External trigger range 0 to 5 volts<br>rising/falling edge<br>impedance selectable 2 k $\Omega$ + 15 pF or 50 $\Omega$<br>minimum recommended amplitude 0.25 volts p-p<br>125 MHz maximum<br>Internal DDS, 0 to 16 MHz, 0.02 Hz resolution |

|                      |                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | Internal clock: 80 MHz source, use Trigger Divisor for Max 16 MHz rep rate                                                                      |
| INTERNAL CLOCK       | Initial accuracy: $\pm 1$ PPM<br>Aging: $< \pm 5$ ppm/1000 hours<br>Jitter: 10ns/s RMS max<br>Lockable to external 10MHz $\pm 10$ PPM           |
| CLOCK INPUT          | 10 MHz, sine or square 0.5 to 5 volts p-p<br>1k $\Omega$ nominal input impedance                                                                |
| CLOCK OUTPUT         | 10 MHz square wave, 3 volts p-p<br>50 $\Omega$ nominal output impedance                                                                         |
| COUNTDOWN            | Provides trigger divide-by-N or N-of-M burst/pulse picker mode, up to 125 MHz external trigger                                                  |
| POWER                | External +12 $\pm 0.25$ volts DC from universal adapter supplied<br>0.5 amps max<br>+24V absolute max voltage input; performance not guaranteed |
| COMMUNICATIONS       | RS-232 standard, 38.4 Kilobaud (configurable up to 460.8 Kilobaud)<br>10/100 Ethernet standard (configurable with serial commands)              |
| PACKAGING            | 4.84" (L) x 4.06" (W) x 1.20" (H) blue extruded aluminum enclosure                                                                              |
| TEMPERATURE          | Specifications apply over 10-40°C ambient<br>Operating range 0 to 50°C<br>Storage range -20 to 80°C                                             |
| CONFORMANCE          | RoHS                                                                                                                                            |
| WARRANTY             | One year limited                                                                                                                                |
| CALIBRATION INTERVAL | One year                                                                                                                                        |
| OPTIONS              | Frame and train engine                                                                                                                          |

### **3 Overview**

The T660 is an embedded digital delay/pulse generator. It accepts a trigger pulse and generates up to four output pulses, with each pulse being individually programmable in delay and width. Triggers may be external, internal, or evoked through the communications interface. Timing has crystal-clock precision with picosecond jitter relative to an external trigger.

Each pulse output is user programmable for delay and width with 10 picosecond resolution, up to a total time (delay + width) of 1 hour (3600 seconds).

#### **3.1 Basic Timing**

A T660 timing sequence begins with a trigger, which can originate from an external source (TRIG connector), internal oscillator, on-board frequency synthesizer, or from a single-shot remote trigger command.

The trigger is gated by the gate, predivider, and burst features, which may allow or disallow the incoming trigger to be recognized and initiate pulse outputs. While the T660 is ready to accept an incoming trigger and that trigger will incite pulse outputs, the unit is considered *armed*. The gate connector may be used to observe armed status.

Once triggered, the T660 is busy until all four pulse output channels have generated user-configured pulses, each pulse having a delay and width. Once complete, an end-of-delay (EOD) period commences while the T660 prepares for another trigger. This sequence, from trigger to the completion of the EOD period, is referred to as a *shot*. Further triggers are inhibited while a shot is in progress.

At any time, the user may initiate a forced end-of-delay (forced EOD); this will interrupt the current shot (if one is in progress) and set all pulse outputs logical low. Some configuration options require that a forced EOD be issued, and issue one automatically.

An EOD sequence normally takes approximately 22.5 nanoseconds, however may take up to 1 microsecond if configuration options have been changed. During this time, incoming triggers will be ignored.

The T660 enforces a minimum delay between the arrival of an external trigger (at the TRIG connector) and the earliest a pulse can be output (from one of the channels, at its connector). This is a fixed time called *insertion delay*, and is factory calibrated to be 21 nanoseconds for all four channels. The T660 automatically performs continuous, real-time temperature compensation to keep this delay stable over changing operating conditions.

All channel timings are measured relative to the assertion of a trigger source plus the insertion delay. This means that a channel configured with a delay of 0 ns will assert 21 ns after an external trigger is received. If a second channel is then configured to 20 ns, it will assert 20 ns after the first channel, or 41 ns since the TRIG connector observed a trigger event. T660 delays are discussed as relative to  $T_0$ , which is defined as the earliest time an output can react after a trigger event.  $T_0$  can be thought of as an internal event that is common across all channels and always occurs 21 ns after a trigger is received. No timing may be negative relative to  $T_0$ .

### 3.2 System Block Diagram

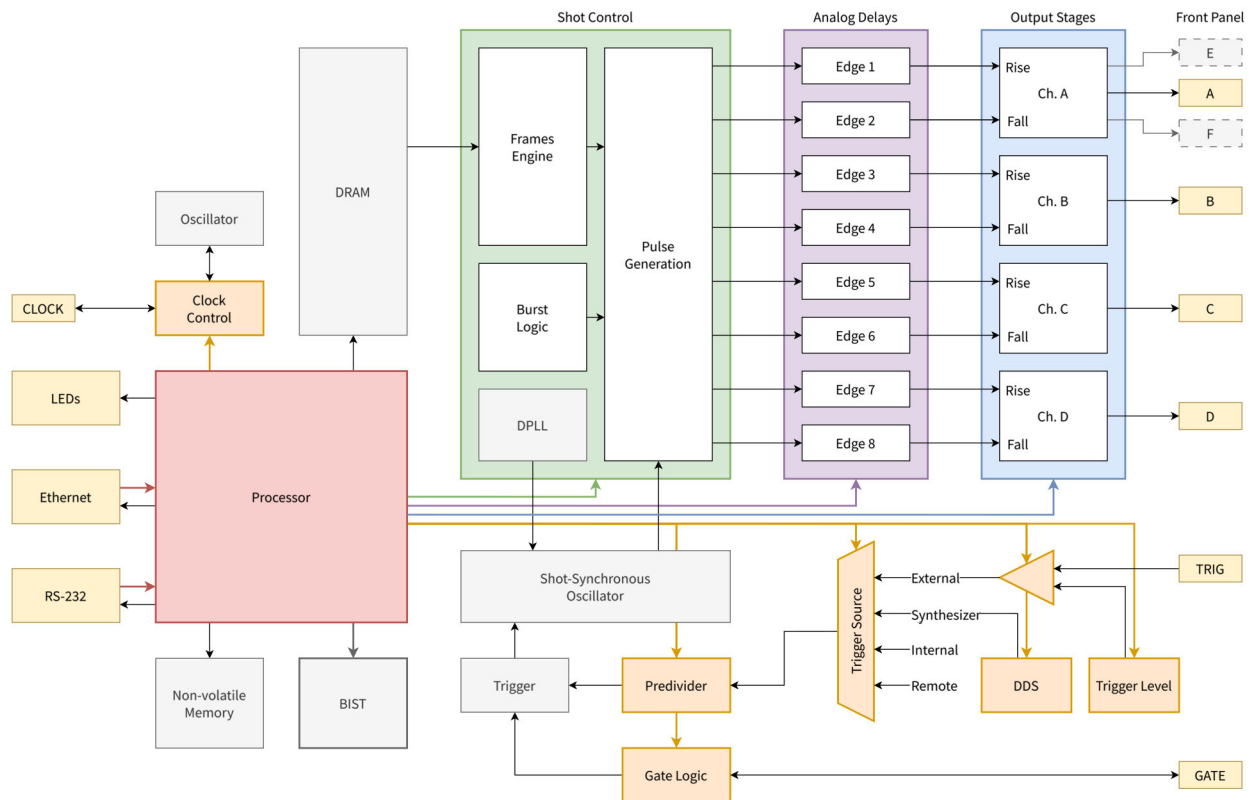


Figure 3-1: T660 System Block Diagram

When a trigger is accepted, a precision instant-start oscillator is initiated. This oscillator is used to count out coarse channel delays. Digitally calibrated analog circuits fine-tune the rising and falling edges of all channel output pulses to 10 picosecond resolution. All timings are agile and accurate on a per-shot basis.

The triggered oscillator is phase-locked to the internal crystal oscillator, which may be optionally locked to an external reference. The digital phase lock system maintains the triggered oscillator phase relative to the asynchronous trigger but ensures long-term accuracy of pulse outputs.

The internal trigger generator is a precision DDS with low jitter and is as accurate as the internal crystal oscillator.

There are four output stages, each programmable for polarity and source termination. They generate clean, fast 50-ohm-source pulses from 0 to 5 Volts into high impedance (Hi-Z) loads, or 50-ohm source pulses from 0 to 2.5 volts into 50 Ohm loads.

The optional *Trains and Frames* features can store a list of timings that can apply to any channels on a per-trigger basis and can schedule multiple pulses on any channel per trigger. This facility can also be used to generate timing sweeps. Refer to sections 5.2 *Trains* and 5.3 *Frames*.

### **3.3 Crystal Oscillator**

The T660 includes an on-board crystal oscillator. It is factory-calibrated to an accuracy of  $\pm 1$  PPM (parts per million) and may be expected to drift less than 2 PPM per year.

The long-delay (millisecond range) jitter performance of the T660 is dominated by the phase noise of the internal crystal oscillator or the equivalent phase noise of a user-provided external reference.

### **3.4 Clock I/O**

A connector is provided which allows the internal oscillator to provide a 10-MHz output or allows an external 10-MHz source to be used as a reference clock. This allows multiple T660s to be synchronized to each or to an external 10-MHz reference.

The output clock signal is a 50  $\Omega$  source-terminated, 10 MHz, 3.3 Volt square wave at CMOS logic levels. The T660 can lock to an externally applied square wave of 10 MHz  $\pm 50$  PPM, 3.3- to 5-Volt positive logic levels, or to a 1 Volt RMS sine wave. When utilized as an input, the clock connector is high impedance. Clock connector functions may also be disabled and the connector left in a high impedance state.

### **3.5 Trigger Input**

The T660 has four trigger sources that may be selected to fire the system: the external trigger (TRIG connector), internal 80-MHz clock, on-board 0-16 MHz frequency synthesizer, and user software trigger.

The selected trigger is first divided by a programmable factor K from 1 to  $2^{32}-1$  in the *predivider*. Once 1 of K triggers is allowed through the predivider, a shot is allowed to begin only if the gate logic allows it. Triggers that occur during a shot (post-predivider) will not interrupt the active shot, but will cause indication of rate errors (see section 3.14 *Indicators*).

The standard external trigger is a positive level, with trigger threshold programmable from +0.25 to +3.30 volts and selectable rising/falling edge. The trigger input may be programmed to be high impedance or a 50-ohm termination to ground. Maximum safe input levels are -0.3 to +5.0 volts.

The maximum supported trigger rate is given by:

$$R = 1 / (D + W + 62.5 \text{ ns})$$

where D + W is the greatest channel sum of programmed delay plus width, and R is limited to 16 MHz max. If a channel is programmed OFF, its time settings are not relevant. If the T660 receives an internal or external trigger while a timing cycle is still busy, that trigger will be ignored. Some additional timing restrictions apply when utilizing *Trains and Frames* features.

An internal 80 MHz clock (exactly 8x the 10 MHz reference clock) may be selected as the trigger source. When it is used, a trigger divisor K must be programmed to keep the trigger rate at or below 16 MHz.

The on-board DDS frequency synthesizer allows triggering at rates from 0 to 16 MHz with 0.02 Hz resolution. The DDS has a period jitter of about 1 part in 20,000, which can

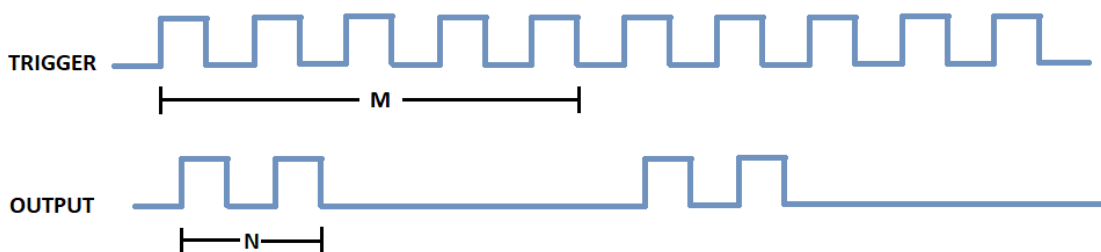
be substantial in absolute terms at low frequencies. Both period jitter and resolution can be improved by keeping the DDS frequency in the 2-10 MHz range and using the trigger predivider to achieve lower trigger rates.

External triggers up to 125 MHz can be accepted, given that the programmed predivisor or the inherent minimum shot duration will restrict the actual triggering rate to a frequency at or below 16 MHz.

### 3.6 Burst Logic

The burst function allows the user to define two integers N and M, each in the range of 1 to  $2^{32}-1$ . When burst mode is enabled, N shots will have pulse outputs, then outputs will remain in their active-low state until M total shots have commenced.

This means that if N=2 and M=5, the T660 will have two active pulses for every 5 triggers, as shown in the following image:



Unlike the predivider, the burst function applies *after* triggers cause a shot, not before, though burst mode also functions for all trigger sources. When triggering with a constant frequency, this difference is typically unimportant, however there are a couple behaviors to keep in mind:

1. Burst counter rate is limited to the maximum *shot* frequency, not trigger frequency, of the T660. If a trigger source is 30 MHz, it must be pre-divided to 15 MHz (with a predivider value of 2) to ensure proper burst operation.
2. Triggers that occur with a counter value between N and M (outputs disabled) still produce shots internally and take the normal amount of time the shot would have taken, just with outputs de-asserted.
3. For frames models only: Shots with their outputs disabled by the burst logic still count as an executed frame to the frames engine, and consequently increment the frame counter, even though outputs were disabled.

Whenever  $N = 0$ ,  $M = 0$ , or  $N > M$ , burst logic is disabled and allows for continuous outputs, meaning that every trigger can fire an output pulse. If  $N=M$ , every trigger shot will be fired.

Burst cycles may "free run", continuously generating N of every M possible cycle. The internal burst logic may be reset by a user command, so that the next trigger will start a new burst of N outputs.

A single burst of N cycles may be started under control of the GATE input or on software command. If GATE mode is set to BURST, the next assertion (logical high

transition) of the GATE input will induce a single burst. Similarly, gate remote mode enables a user-fired burst, started by the “gate fire” commands.

### **3.7 Pulse Outputs**

Four pulse outputs are provided: channels A, B, C, and D. Outputs are +5-volt CMOS levels with a programmable 50-ohm or low (Lo-Z) source impedance. If the outputs are configured to be 50-ohm source terminated, they can drive 5 volts into a non-terminating load, or 2.5 volts into a 50-ohm load. With a low source impedance, the outputs can drive >4 volts into a non-terminating load and into a 50-ohm load.

Each output is programmable for pulse delay and width relative to the trigger. If an output is programmed for delay D, the actual output pulse will occur at D + 21 nanoseconds relative to an external trigger, where 21 nanoseconds is the insertion delay of the T660. Pulse outputs are active-high by default and may be programmed to operate active-low.

Changing output delays requires time for the analog circuitry to settle, during which time all triggers will be ignored; this occurs during an EOD (end-of-delay) event. While channel outputs are being reconfigured, EOD is extended to approximately 1 microsecond.

Because their outputs can be source terminated, channels may drive a 50-ohm coaxial cable any distance into any termination impedance without significant reflection problems. For example, a 50-ohm source termination with a 100-ohm end termination will provide a clean 0-to-+3.3-volt logic swing, and a high-impedance (such as CMOS) load will swing 5 volts. External passive components can be used to convert to ECL or PECL levels.

### **3.8 Gate I/O**

A GATE coaxial connector is provided. It is pulled up to +5 volts through a 1K resistor or may, under software control, be terminated by 50 ohms to ground when used as an input. The GATE connector may be programmed as an input or as an output.

As an input, the GATE connector may be programmed to allow triggers, start bursts, or provide AUX output functionality (Trains and Frames models only).

As an output, the GATE connector may be programmed to indicate when the unit is armed and the next received trigger will cause a shot with outputs to commence.

In all selectable modes, both input and output, the GATE connector may be configured to be logic-high or logic-low.

### **3.9 *Saved Configurations***

Users may program the T660 as desired and then save its configuration to nonvolatile flash memory with a command. The saved configuration is also loaded and applied at powerup, allowing the T660 to resume operation without any serial commands. The saved configuration may also be loaded and applied at any time with a command.

The T660 has four independently saved configurations:

- Network (IPv4 configuration only)
- Serial (Baud rate only)
- Clock (VCXO trim only)
- Global (all other non-session configurable settings)

Frames are not saved to nonvolatile memory.

The global configuration is typically the one most relevant to device operation, since it includes channel, timing, trigger, predivider, burst, clock (non-trim), trains, and other miscellaneous configurations such as P500-style commands channel relativity and T560-style commands autoinstall and verbose modes.

### **3.10 *Power-up Sequence***

The powerup sequence takes approximately 10 seconds. During this time, channel outputs are electrically low and terminations are high-Z. During power-on, the network, serial, clock, and global configurations are loaded from nonvolatile memory and applied. Ethernet connectivity may require additional time to initialize depending on saved network configuration.

If the saved configuration programs channels to be inverted polarity, those outputs will transition from low to high when the initial powerup sequence is complete.

### **3.11 *Communications and Control***

All versions of the T660 are equipped with a RS-232 interface (38400 baud default) and a RJ45 10/100 Ethernet interface. Multiple simultaneous ethernet and RS-232 communications are fully supported. See section 4.2 *Session Interfaces* for details on how to use these interfaces.

The T660-10 does not have Ethernet connectivity, though does include an additional header for communication and configuration. See the *T660 Manual -10 Addendum* (available upon request) for details regarding this interface.

#### **3.11.1 *Hostname***

The T660 uses a host name of the form “T660-XXXXX”, where “XXXXX” is the serial number of the unit padded with zeroes on the left to five digits, for example “T660-00012” for serial number twelve.

The hostname is broadcast via mDNS and to the DHCP server upon connection, if enabled. This means that if the DHCP suffix is, for example, “localnet”, a T660 with serial number 38 may be accessed by two domain names: “T660-00038.local” (mDNS) and “T660-00038.localnet” (DHCP).

### 3.11.2 IPv4

By default, the T660 uses a DHCP client to request an IP address from the network's DHCP server. This may be changed to a static IP address using serial commands; this change is stored to non-volatile memory and will persist after being applied.

### 3.11.3 IPv6

The T660 supports dual stack networks and includes IPv6 support. This support extends to mDNS and DHCP address lookups. IPv6 address and configuration is not accessible via user commands, so IPv6 utilization is only useful via domain name lookups. Therefore, it is suggested to use the mDNS name ("T660-00038.local") to interface with a T660 using IPv6.

## 3.12 Connectors

Standard logic-level connectors are right-angle SMBs. A three-foot SMB-to-BNC cable is available as Highland part number J53-1; two of these are included with every T660 purchase.

A miniature 2.5-mm stereo phone jack is used for the RS-232 serial interface; a mating connector and cable is available, terminated with a female D9 connector, and may be plugged into the serial port of a standard PC. The RS-232 cable assembly is Highland part number T565-1, and one is included with every purchase.

A pinout and diagram of the serial connector cable is provided below.

| Name | Description           | 2.5mm TRS | DB9 Female |
|------|-----------------------|-----------|------------|
| RXD  | Serial data to T660   | Tip       | D9-3       |
| TXD  | Serial data from T660 | Ring      | D9-2       |
| GND  | Common ground         | Ground    | D9-5       |

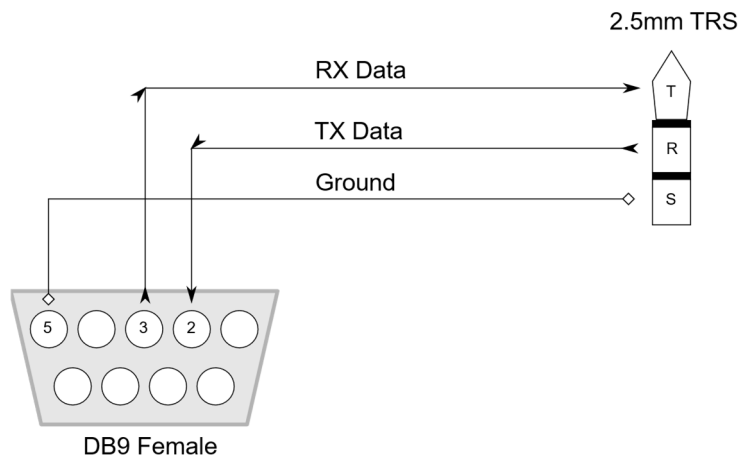


Figure 3-2: 2.5-mm Stereo Audio Jack to D9 cable connections

### **3.13 Power Input**

The T660 requires +12 volts DC at 0.5 amps max. A wall-plug universal power supply is furnished with the evaluation kit, or users may supply +11.75 to +12.25 VDC power. A Highland J12 +12 Volts wall-wart power supply and an international AC plug adapter kit is provided with every purchase.

The standard power connector is a 2.1 x 5 mm coaxial power type, center positive.

The T660 power input will safely withstand +24 Volts but may not operate to specifications. DC power should be nominally 12 Volts.

### **3.14 Indicators**

The T660 has two multi-color LED indicators, labeled TRIG and PWR.

The PWR LED is normally green and blinks off every 1.5 seconds to indicate that the unit is operating correctly, while the TRIG LED maintains a color to indicate a status and is normally off. Each color represents a system status, and each status has a priority. A status of higher priority will be shown instead of one of lower priority if multiple conditions are true. A status will be lit for a minimum of 0.1 seconds to ensure no quick conditions are visually missed.

The statuses for each LED are listed below, listed from top to bottom from low to high priority respectively.

| <b>Color</b> | <b>PWR LED</b>                                                                         |
|--------------|----------------------------------------------------------------------------------------|
| Green blink  | Blinks off every 1.5 seconds to indicate the unit is responsive with no system errors. |
| Blue blink   | Blinks with communication activity                                                     |
| Red          | System error(s) are present                                                            |
| Magenta      | Both Blue and Red conditions are true.                                                 |

| <b>Color</b> | <b>TRIG LED</b>              |
|--------------|------------------------------|
| Off          | No triggers received         |
| Blue blink   | Blinks with received Trigger |
|              |                              |
| Magenta      | Trigger rate error           |

## **4 Configuration**

### **4.1 Introduction**

The T660 is configured using ASCII commands from two independent, feature-complete command sets. These two command sets provide style compatibility with other Highland Technology products:

- P500-style (compatible with the P400 and P500 Digital Delay Generators)
- T560-style (compatible with the T560 and T564 Digital Delay Generators)

Consult the *T660 Programming Guide* for details on how to use these command sets and full command lists for each.

To take full advantage of all the T660's features, the P500-style command set must be used; all T560-style commands have a P500-style equivalent, but the reverse is not true. However, all functionalities offered by the T560 and T564 are incorporated into the T560-style command set, along with many new T660 features.

### **4.2 Session Interfaces**

A single T660 accepts and responds to ASCII commands from any combination of physical ports and session interfaces simultaneously. Active command sessions may be initiated over any physical port or session interface and are all treated with equal priority. The number of command sessions initiated over Ethernet is unlimited, so care should be taken to ensure that the number of active Ethernet connections remains reasonable to avoid device memory issues.

There are four session interfaces which each accept and service command sessions. A comparison of these interfaces is provided below:

| <b>Session Interface</b> | <b>Physical Port</b> | <b>Max. Connections</b> | <b>Default Mode</b> | <b>Session Types</b> |
|--------------------------|----------------------|-------------------------|---------------------|----------------------|
| Serial                   | RS-232               | 1                       | T560-style          | All                  |
| TCP (Port 2000)          | Ethernet             | Unlimited               | T560-style          | T560-style only      |
| TCP (Port 2001)          | Ethernet             | Unlimited               | P500-style          | P500-style only      |
| Isochronous <sup>1</sup> | Embedded             | 1                       | T560-style          | All                  |

<sup>1</sup>Available on T660-10 units only.

The behavior of command sessions is identical regardless of the physical or digital interface used to create or access them, and their behavior is consistent across all interfaces. This includes basic text formatting (newline and whitespace compatibility is identical across all interfaces), and command interpretation (commands are interpreted identically regardless of the session's interface of origin), for example.

For evaluation, serial commands may be typed using any common serial communications program, for example TeraTerm or Putty.

#### **4.2.1 T660 vs T560 IP Configuration**

Previously for Ethernet connectivity the T560 and T564 could use the Lantronix Xport module as the TCP-IP interface and webpage server. The Lantronix “XportInstaller” PC utility is no longer supported to function with the T660 as a method to locate and re-assigning an IP address.

The following steps specify how to access a T660 through DHCP Hostname, or Serially through RS-232:

1. *Establish a DHCP session with the T660 using a Terminal program (eg. Tera Term or puTTY) through the host name t660-xxxxx.local w/ port 2000, where xxxxx is the T660’s serial numbers padded with leading zeros.*
2. *Assign a fixed IP address to the T660 that is compatible with the system’s network using the NEtwork ADdress <x.x.x.x> command, where x.x.x.x is the desired IP address. Note that since the T660 doesn’t have a built in web server, it won’t be accessible by entering 192.168.254.183 in a browser as done in the past with the T560.*
3. *Commit the new T660 IP address using the NEtwork APply command.*
4. *Reboot the T660 by power cycling or using the RSet command.*
5. *Verify the T660’s IP address by Pinging or querying with the NEtwork ADdress command.*
6. *If unable to connect through a DHCP session, the commands from step 2 through 5 can also be applied using a serial RS-232 communication session. Refer to Section 4.2.2 for RS-232 connection parameters.*

Note that entering the T660’s IP address in a browser will bring up the firmware upgrade webpage. A telnet terminal program or raw-TCP client is required to send serial commands or queries.

### **4.2.2 RS-232 Interface**

The serial interface can be accessed via the RS-232 serial port. Upon power-up, the serial interface operates using a T560-style session. The default baud rate is 38400 baud, however this may be changed using commands.

The RS-232 port uses the following connection parameters:

|                      |                                                                         |
|----------------------|-------------------------------------------------------------------------|
| Supported Baud Rates | 9600<br>19200<br>38400 (default)<br>57600<br>115200<br>230400<br>460800 |
| Data Bits            | 8                                                                       |
| Stop Bits            | 1                                                                       |
| Parity               | None                                                                    |

### **4.2.3 TCP Interfaces**

Ports 2000 and 2001 are used for the ASCII serial commands described in the serial command reference. These are raw TCP sockets. Users can use a telnet or raw-TCP client to send serial commands or queries. Note, however, that the first line of a telnet session may always return as an error due to invisible character transmission as part of the telnet protocol.

The T660-10 does not have Ethernet connectivity; the isochronous interface is primarily used for configuration and communication.

### **4.2.4 Isochronous Interface**

The Isochronous interface is only available on T660-10 units. For details, refer to the *T660 Manual -10 Addendum* (available upon request).

### 4.3 *Staging Sets*

To simplify the interactions between the T560-style and P500-style command sets, the T660 introduces the concept of *staging sets*. A staging set is a set of configurations that contains all channel and timing configuration. Specifically, this includes:

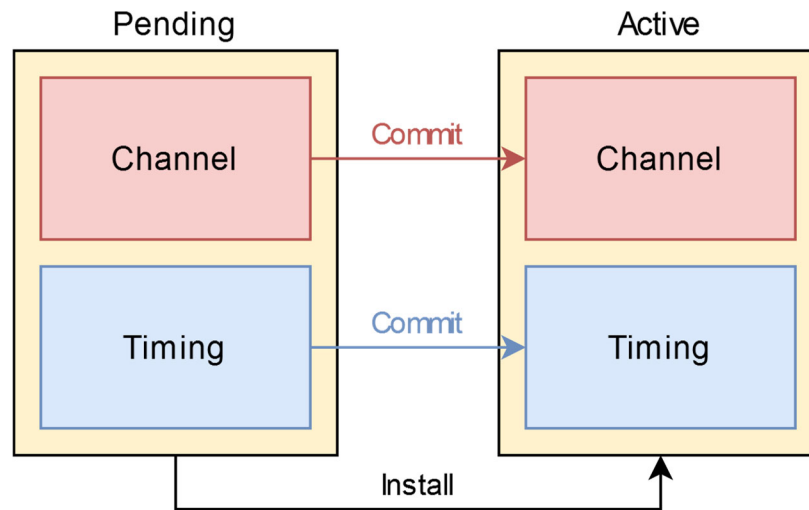
- Channel configuration (4x; 1 per channel)
  - Enable
  - Polarity
  - Termination
- Timing configuration (4x; 1 per channel)
  - Delay
  - Width
- Trains configuration (Trains feature required)
  - Train count
  - Train spacing

The T660 has three staging sets, also referred to as staging configurations: *active*, *queued*, and *pending*. The *active* staging set represents the current hardware configuration, the *queued* (Trains and Frames required) set represents configuration that will be loaded into hardware (“applied”) at the next end-of-delay event, and the *pending* set is software-only configuration that has not been installed or queued yet.

Configuration is applied in sequence from pending, queued, then to active (hardware) depending on events and user-initiated actions.

Staging sets are a complicated topic, so two explanations are provided: a standard explanation (applicable to the T660-1), and one with the *Trains and Frames* features considered (applicable to the T660-2).

### 4.3.1 Standard Explanation



Channel and timing commands modify the pending configuration, which is then applied to hardware through a commit or install operation. The active configuration is a read-only copy of what the hardware is currently configured to do.

The user can issue two relevant operations: *install* and *commit*. An *install* copies the pending configuration to the active configuration and applies this configuration to hardware. This process also initiates an end-of-delay (EOD) event, which interrupts any currently running shot. A *commit* is like an install operation, but only applies either the channel or timing configuration, depending on the command used.

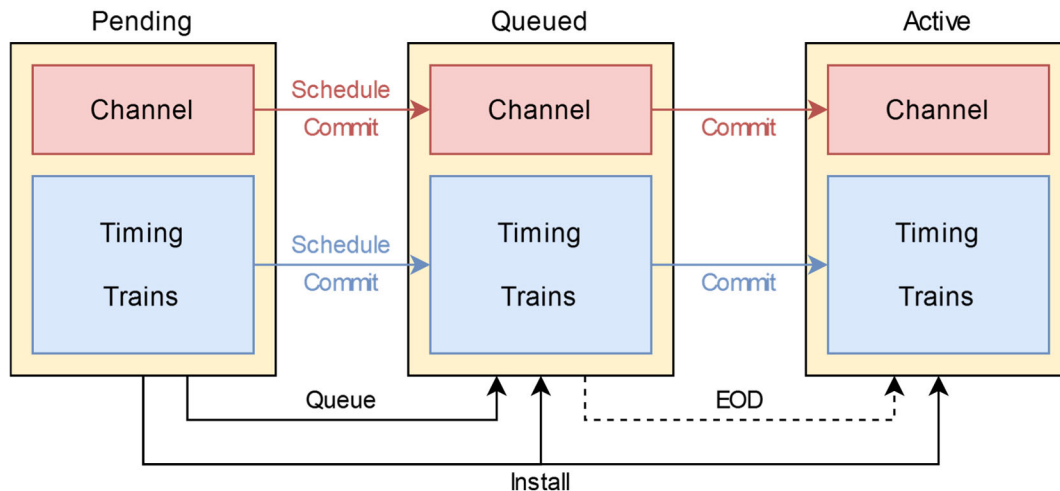
Until an install or commit is issued, changes made to the pending configuration will not affect hardware nor the active configuration.

An install operation can be issued manually with a command or using the T560-style autoinstall feature, which issues an install when necessary at the end of a T560-style command line. The commit operation is only available using P500-style commands.

### 4.3.2 Trains and Frames Explanation

The Trains and Frames feature allows for updating pulse output configuration without interruption of a running shot. Using the *queue* and *schedule* operations guarantees that a running shot will not be interrupted.

A staging set holds the same information contained within a frame: channel, timing, and trains configurations.



Channel, timing, and trains commands modify the pending configuration, which is then applied to hardware via a user-initiated *queue* or *install*. The user may also issue a *schedule* or *commit*, which act as partial configuration equivalents to these two operations.

The active configuration is a read-only copy of what the hardware is currently configured to do, and the queued configuration is applied to hardware at the end of a shot, whenever it has been changed with a *queue* or *schedule* operation.

Actions and events that affect staging configurations are as follows:

- **End-of-Delay (EOD)** – This event occurs at the end of every shot or may be initiated via a user command at any time. When the queued staging set has changes waiting to be applied to the active set, EOD will be extended and the queued changes applied to hardware (the active staging set).
- **Queue** – Copy the pending set to the queued set. The queued configuration will be applied to the active set at the next EOD.
- **Install** – Perform a queue, then force an end-of-delay. This interrupts any currently running shot and applies the pending set to both the queued and active sets.
- **Schedule** – Copy a partial configuration (depending on command) to the queued set. The queued configuration will be applied (to the active set) at the next EOD. This can be thought of as a partial queue.
- **Commit** – Issue a schedule, then force an end-of-delay. This can be thought of as a partial install.

Unlike the active and queued staging sets, which can only be modified by the actions/events listed above, the pending set may be modified at any time using the various channel, timing, and trains commands. Until an install, queue, commit, or schedule is issued, the pending configuration will not take effect.

When an EOD occurs and the queued configuration is about to be applied, the EOD event will take approximately 1 microsecond instead of its normal approximately 22.5 nanoseconds, during which time the active configuration and hardware will be updated; an install is performed synchronously to the shot. This only occurs when the queued set has been modified since the last install.

## **5   *Trains and Frames Features***

An advanced version of the T660 (T660-2), provides *Trains and Frames* capabilities. Fundamentally, this feature adds the ability to:

1. Automatically update channel/timing configuration both online and offline without interrupting shots (Frames and Queue commands)
2. Output multiple pulses per shot, all synchronous to the initial trigger (Trains)
3. Utilize the gate connector to provide insight into device operation (Auxiliary Output)
4. Store multiple timing, channel, and trains configurations, and reload them quickly with commands (Install frame and Queue frame commands)

The Trains feature allows a single trigger to invoke a sequence of as many as  $2^{32}$  sets of programmed delays.

The Frames feature allows users to pre-load a list of up to 8192 delay and width settings and enable the unit to rapidly reload these settings between shots, without real-time per-shot serial commands.

All Trains and Frames features may be used separately or together.

### **5.1   *Auxiliary Output***

The gate connector may be configured as an auxiliary output. This output generates signals that do not affect gate operation, despite utilizing the gate connector.

When the gate connector is configured to be in auxiliary output mode, gate functionality behaves as though the gate is disabled. The auxiliary output is affected by gate polarity configuration just like any other gate mode.

There are two auxiliary output modes: *update* and *trigger* mode. In update mode, the auxiliary output mirrors the perceived trigger input; if trigger polarity is negative, the output will be inverted. This is useful for observing, for example, internal triggers.

In update mode, the auxiliary output outputs a pulse of configurable width to indicate an install has just completed. The output pulse may be up to 3 milliseconds long and is quantized to 20 nanoseconds.

**Caution:** the auxiliary output is not calibrated nor guaranteed to be synchronous to the trigger, gate, and pulse outputs. It may have up substantial skew from trigger to output and increased jitter. Therefore, this output should be used for diagnostics, debugging, and status to external equipment only.

## 5.2 Trains

The T660 may be programmed to produce pulse trains; to produce multiple output pulses starting with a single trigger. The pulse outputs in the example below were generated from a single trigger.

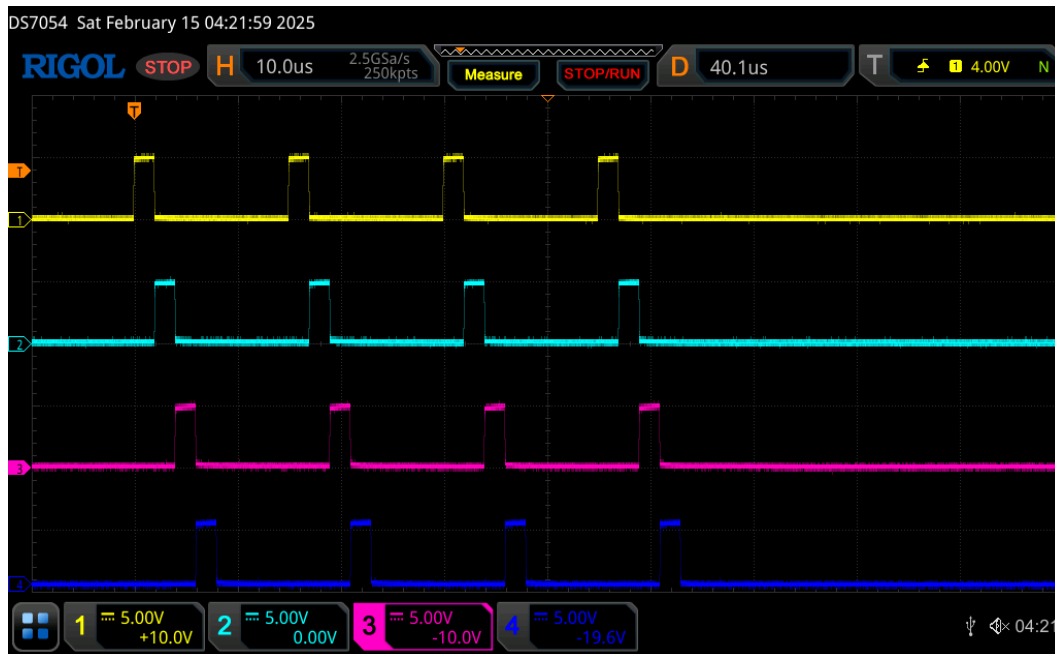


Figure 5-1: Channels A-D outputs (Ch. 1-4) with a train count of 3.

The first set of A/B/C/D outputs are the usual ones, as programmed by the standard channel delay and width commands. After these are finished, *train count* additional pulses are generated on each channel, each separated by a configured time, *train spacing*. The sequence above was invoked with a train count of 3 and a train spacing of 15 microseconds. Up to  $2^{32}-1$  pulses can be generated by the train facility, with spacing from 80 ns to 10 seconds.

Train spacing is quantized to intervals of 20 ns. Additionally, train spacing must be set such that for any given channel, there must be a minimum dead time (logical low) of 40 ns minimum to ensure no skipped pulses. The dead time is specified as the duration from falling edge to rising edge (or rising edge to falling edge if the channel polarity is negative).

After all pulses are over, the standard EOD (end-of-delay) system reset occurs.

Train parameters must be installed before becoming active, using install or queue commands, or their autoinstall equivalents. Train parameters are saved in Frames.

If any channel is programmed for a delay below 20 ns in Train mode, only the first, output pulse will appear. This is a feature that allows "train start" output pulses to be

generated. In the example below, the channel A delay is set to zero, so it makes only one initial pulse when trains are active.

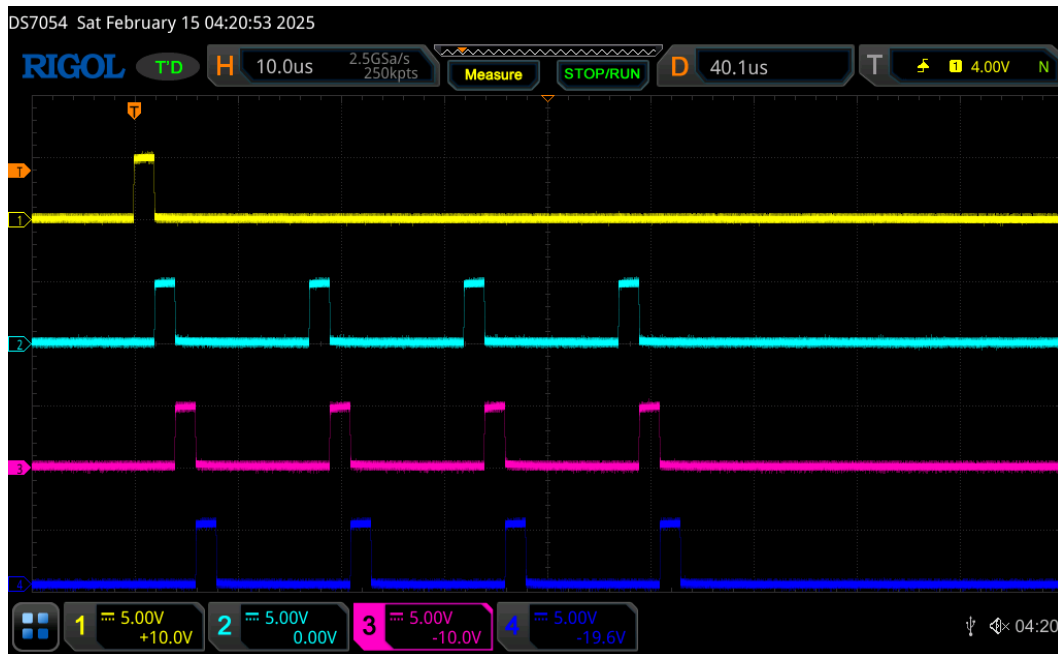


Figure 5-2: Channels A-D outputs with channel A set to 0 ns delay.

### 5.3 Frames

The T660 allows a schedule of multiple delay setups to be stored in local volatile memory, with new settings loaded rapidly after each trigger/delay sequence.

Delay setups are compiled and stored in *frames*. Each frame includes channel, timing, and trains configurations; this is identical to the information contained within a staging set (see section 4.3 *Staging Sets* for additional details). Users may configure these settings and store this setup to a numbered frame. The T660 can store up to 8192 frames.

Frames' primary use-case is to be used with the frames engine, detailed in the following sections.

Frames can also be used without the frames engine: if frames have been stored but the frames engine is inactive, *install frame* and *queue frame* commands may be issued to load or apply a specified frame's configuration into hardware. This allows the frame buffer to be used to store saved timing setups, any of which can be recalled quickly with a single command. These commands are also useful for testing frames engine configurations.

### 5.4 Frames Engine

The *frames engine* can be thought of as a real-time co-processor that applies hardware configurations at the end of a shot. It executes independently of commands and allows a programmable number of shot configurations to be loaded using parameters extracted from sequential frames. While the frames engine is running, it takes control over the normally command-configured channel, timing, and trains configurations. During this time, all frames and frames engine parameters may not be changed via commands.

While active, the frames engine loads a frame into hardware, waits it to execute (on receipt of a valid trigger), and applies the next sequential frame into hardware at EOD. The engine loads frames sequentially by frame index (0-8191), beginning from a start index to an end index and may be configured to loop over this range a fixed or indefinite number of times.

The frames engine does not initiate shots, nor does it affect other sub-systems of the T660, such as the trigger, predivider, burst, and gate subsystems. This means that the burst subsystem may be used while the frames engine is running, though this can produce unintuitive results.

**Note:** There are limitations to frames engine performance. Trigger rate may not exceed 1 MHz due to the required 1 microsecond hardware reconfiguration period between triggers. Violating this limit may cause triggers to be missed.

### 5.4.1 Operation

The frames engine has three states:

- OFF: The frames engine is disabled. DDG operates normally.
- RUN: New frames are loaded at each end-of-delay (EOD).
- DONE: The last frame has completed, and triggers are inhibited.

It also has the following configurable parameters:

- FA: 0-8191. Starting frame index. This is the first frame to be loaded, and the starting index of loops.
- FB: 0-8191. Last frame index. This is the last frame to be loaded, and the ending index of loops.
- FC: 0-65534 (or infinite). Loop count. The number of times to load frames FA through FB. 0 means that (FB-FA+1) frames will be loaded, and 1 means one additional loop will be performed (2\*(FB-FA+1) total). A value of 65535 loops forever.

At power-on, the frames engine is OFF by default, until started by a command. When started, the engine moves into the RUN state and operates until it has applied a configured number of frames (or is turned off with a command). It then moves into the DONE state, where it prevents further triggers until turned OFF with a command.

The frames engine may be started and deactivated using engine start and engine stop commands, present in both the T560-style and P500-style command sets.

An *engine start* command loads the start (FA) frame into the hardware and queues the next sequential frame to be installed at the next EOD. Subsequent triggers use the timings extracted from sequential frames. After the last (FB) frame has been initiated by a trigger, further triggers are disabled until an engine start or stop command is issued, during which time the engine will be in the DONE state. The frames engine may only be started if FB is greater than (not equal to) FA.

An *engine stop* command returns the T660 to its normal operating mode. Triggers will be re-enabled, and the active channel, timing, and trains configuration that was applied before the frames engine was activated is re-installed, causing all timing sets to be over-written with this configuration. Triggers may be disabled prior to running an engine stop command if this sequence produces undesired or spurious outputs.

Accessible through commands, a read-only value FN represents the number of frames loaded into hardware by the frames engine. This value is automatically updated by the frames engine upon EOD, and is clearable by command. FN count is preserved across frames engine starts and stops and is only cleared by command or power off.

When the frames engine is started, an additional install will be performed, meaning that every time an engine start command is run, the FN count will increment by 2.

### 5.4.2 Loops

If the FC parameter is zero, a command will load frames FA through FB into the hardware then stop, responding to (FB-FA+1) triggers.

If the FC loop parameter is in the range of 1 to 65534, that sequence will be repeated FC times, responding to (FB-FA+1) \* (FC+1) triggers and shots; about 536 million triggers in the extreme case. If FC is set to 65535 and the frames engine is started, the frames engine will loop forever, executing frames FA through FB until terminated by an engine stop command.

In T560-style command sessions, to speed up configuration of frames, it is strongly recommended that autoinstall be off (zero): it is not necessary to issue an install command before saving a frame, and installs add additional command execution time. Similarly, in P500-style command sessions, it is recommended to use commands that do not automatically execute an install or commit as part of their operation.

### 5.4.3 Other Uses for Frames

Since channel enable (on/off) and polarity states are saved in frames, it is possible to have channels selectively fire on a frame-by-frame basis. One can also configure a channel as off while still changing its polarity on a per-frame basis, effectively creating a static TTL output that is controllable per frame. Note, however, that channel polarity and termination are not perfectly synchronous to the end of a shot; these settings may be applied up to 1 microsecond after the start of EOD.

### 5.4.4 Example

In the example below, one of the following sets of serial commands is sent to the T660:

| T560-Style Commands                                                                                                                                                                                               | P500-Style Commands                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Load Default<br>TDiv 80000<br>BSet Off<br>CSet Off; DSet Off<br>ADelay 0<br>AWidth 100u; FRame 1<br>AWidth 200u; FRame 2<br>AWidth 300u; FRame 3<br>AWidth 400u; FRame 4<br>FA 1; FB 4<br>FRame GO<br>TRigger INT | MEMory:LOAD:DEFault<br>TRIGger:PREDiv 80000<br>CHAN:MODE B, OFF<br>CHAN:MODE C, OFF; MODE D, OFF<br>TIME:DELay1 0<br>TIME:QUEue2 100u; :TFRame:STORe 1<br>TIME:QUEue2 200u; :TFRame:STORe 2<br>TIME:QUEue2 300u; :TFRame:STORe 3<br>TIME:QUEue2 400u; :TFRame:STORe 4<br>TFRame:LOOP:FIRST 1; LAST 4<br>TFRame:STArt<br>TRIG:SOURce INT |

This will result in four triggers being fired at a 1 kHz rate, producing pulses of 100, 200, 300, and 400 microseconds on the channel A output connector. The load default command clears the frame loop counter FC, so frames 1 through 4 are scanned only once. The FN counter is incremented by five, the number of triggers plus one.

The resulting waveform is shown below, with the A output displayed.

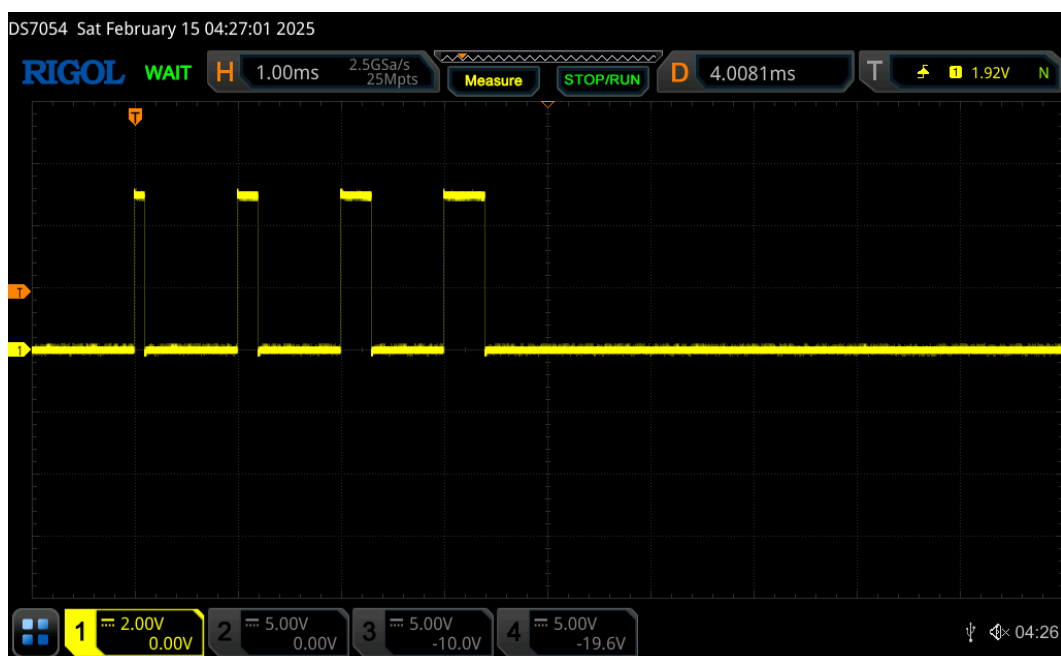


Figure 5-3: Frame output example

Starting the frames engine again will repeat this sequence.

If FC is set to 1 and the frames engine re-started, the four programmed frames will be executed twice, accepting a total of 8 triggers, as shown below.

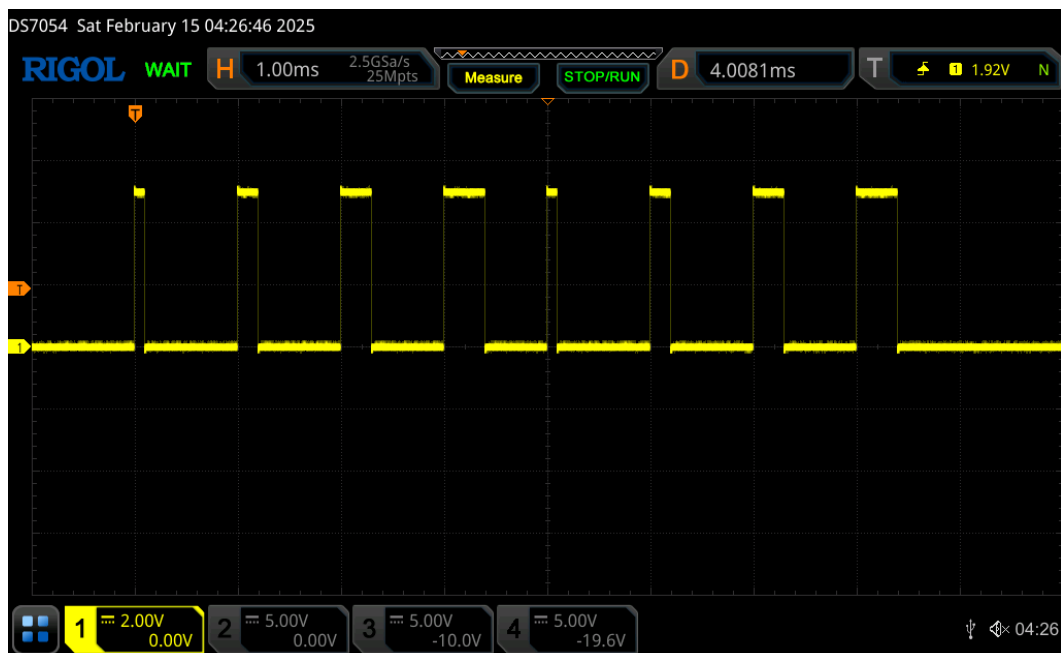


Figure 5-4: Repeating FRAME example

## **6 Application Notes**

### **6.1 Command Execution Time**

Most simple commands execute in hundreds of microseconds, and their real-time execution rate is dominated by the 38.4 kilobaud (3840 characters/second) serial communications rate. Short-form commands reduce communications overhead. Long responses may be baud rate limited, with the T560-style status report taking as long as 500 milliseconds at the default baud rate.

Command execution times are usually dominated by the default 38.4 kilobaud communications rate. Since command execution times include the time taken to transmit a response, a command executed via the serial port may take much longer than the same command over an Ethernet session.

### **6.2 Applying Configuration**

When delay/width settings are changed via an install, commit, or force-EOD command or the trigger, burst, or gate parameters are changed, the firmware will immediately force the end-of-delay reset state, which will abort any timing cycles (shots) currently in progress. EOD will be asserted for about 1 microsecond, after which triggering will be re-enabled.

If aborting timing cycles is undesirable, one can disable triggers, wait until any possible timing cycle has finished, then perform the desired operation. Models with the *Trains and Frames* feature (the T660-2) also feature a queue operation, which allows delays to be reprogrammed without disrupting ongoing shots.

### **6.3 Jitter Notes**

Jitter is defined as the standard deviation of delay. It is the shot-to-shot time uncertainty from the external trigger to any output's rising or falling edge, or the uncertainty between edges of channel outputs. Jitter is measured in RMS picoseconds. Visual peak-to-peak jitter is roughly 5 times that of RMS.

"Jitter" is usually accepted to indicate time variance as observed over an interval of 0.1 seconds, with the term "wander" used to describe slower changes of delay. Wander thus encompasses changes in delay driven by temperature changes and other slow effects. The T660 jitter specs are valid for observation periods up to 10 seconds in the absence of radical temperature changes. Note that coaxial cable propagation delay can change considerably with temperature and can contribute to observed timing variance.

Uncorrelated jitters add trigonometrically, as the square root of the sum of the squares of all jitter contributors.

Jitter can be difficult to measure. The trigger input to the T660 must be clean and fast (< 2 ns risetime) and the measuring instrument must have a jitter noise floor well below that of the T660. Most oscilloscopes and counters are not capable of resolving T660 jitter performance, especially so for longer delays. For example, a Tektronix 11801C sampling oscilloscope (or the newer DSA8200 without the optional phase lock module) has a short-delay jitter well below that of the T660 but has added jitter on the order of 20

microseconds per second of delay, whereas the T660 starts with a greater basic jitter but typically adds about 4 ns of jitter per second of delay.

For the lowest jitter from an external trigger, the T660 trigger level should be set to the steepest part of the input edge, typically 1/3 to 1/2 of the peak amplitude.

Jitter is a function of the generated time delays. Very short delays have a baseline jitter that depends on fundamental triggered-oscillator phase noise. After about 500 ns, the DSP stabilization loop becomes active and disciplines the triggered oscillator, limiting its jitter accumulation.

Long delays, in the milliseconds range, become dominated by the phase noise of the internal crystal oscillator, typically about 4 ns per second of delay. Long-delay effects are zero relative to a user-provided 10 MHz reference clock.

Jitter between successive triggers, referred to as "period jitter", depends on the quality of the trigger source. The internal DDS trigger synthesizer has jitter typically about one part in 20,000 of the trigger period. DDS jitter is best if its frequency is in the 2-10 MHz range, where the period jitter, measured at a channel output, is typically about 25 ps RMS. For lowest DDS jitter at lower rates, keep the DDS synthesizer frequency in this range and use a trigger divisor to get lower trigger rates.

## 7 Dimensions and Mounting

T660 mechanical dimensions are shown below. The unit may be mounted with four 4-4-machine screws from below the box.



**CAUTION:** Mounting screws may not penetrate more than 0.160 inches (4 mm) into the T660 enclosure.

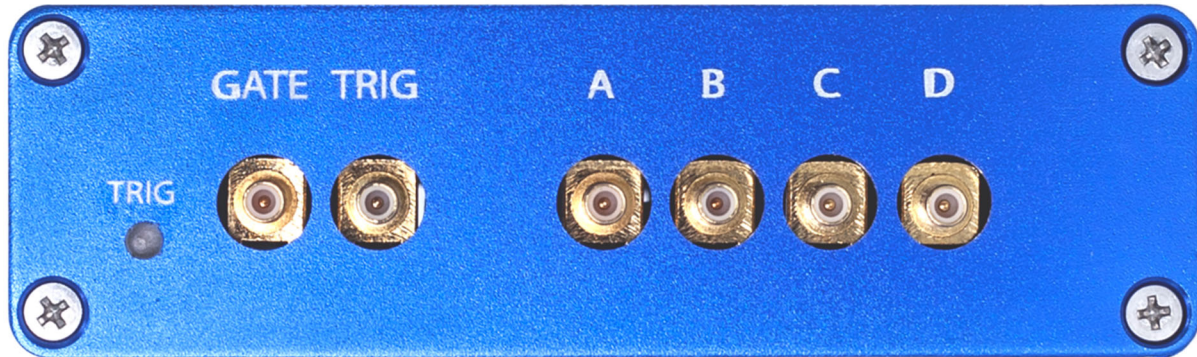


Figure 7-1: T660 Front Panel



Figure 7-2: T660 Rear Panel

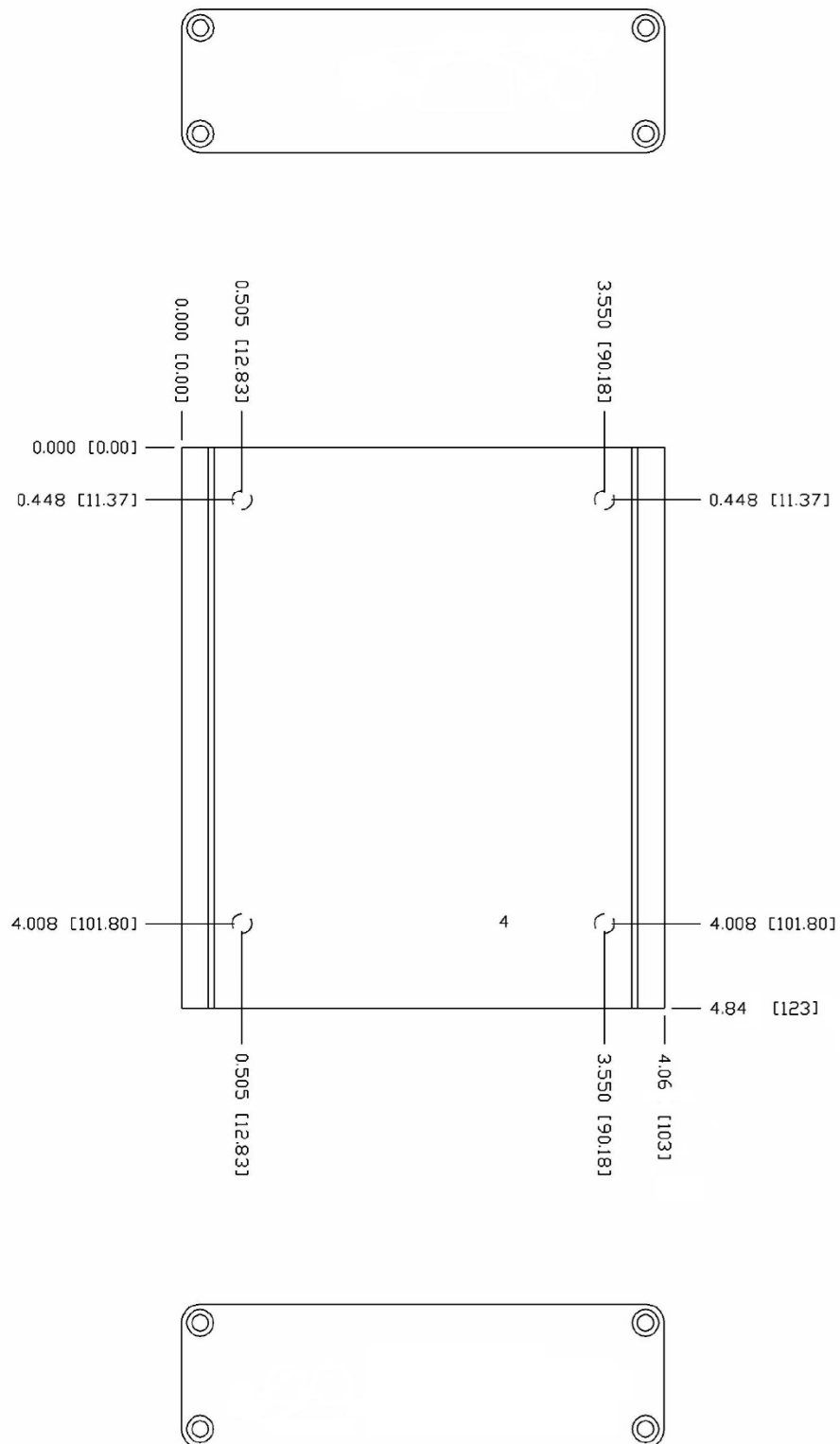


Figure 7-3: T660 enclosure and mounting locations

## **8 Versions**

- T660-1: 4-channel compact digital delay and pulse generator
- T660-2: 4-channel compact digital delay and pulse generator with TRAINS and FRAMES capability
- T660-10: Special OEM version, 4/6-channel compact digital delay and pulse generator with TRAINS and FRAMES capability

## **9 Customization**

For more information about the T660-10 Special OEM version, a T660 Manual -10 Addendum is available upon request.

Consult factory for information about additional custom versions.

## **10 Revision History**

### **10.1 Hardware Revision History**

|            |                                                                                    |
|------------|------------------------------------------------------------------------------------|
| Revision F | August 2024<br>Improved Manufacturability<br>Functionally equivalent to Revision E |
| Revision E | May 2024<br>Initial release                                                        |

### **10.2 Software Revision History**

|              |                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28E660-1-1.7 | June 2025<br>Improved T564-10 isochronous interface compatibility for T660-10 units and removed the welcome banner from T560-style TCP connections.             |
| 28E660-1-1.6 | April 2025<br>Improved T564-10 isochronous interface compatibility for T660-10 units.                                                                           |
| 28E660-1-1.5 | February 2025<br>Added serial baud rate commands, additional error flags, improved DDS and frequency counter performance, and added P500-style command support. |
| 28E660-1-1.3 | November 2024<br>Added support for Auxiliary Output functionality and overlapping trains. All T560-style help commands were deprecated.                         |
| 28E660-1-1.2 | August 2024<br>Improved internal temperature compensation.                                                                                                      |
| 28E660-1-1.1 | June 2024<br>Bug fixes for channel outputs during configuration changes and gate burst functionality while burst is off.                                        |
| 28E660-1-1.0 | May 2024<br>Initial production release.                                                                                                                         |

## ***11 Accessories***

- J12-1: 12 volt power supply (1 included with purchase)
- J53-1: 3' SMB to BNC cable (2 included with purchase)
- J53-2: 6" SMB to BNC cable
- P10-1: 19" rack mount shelf (four t-boxes per rack)
- T565-1: RS-232 cable (1 included with purchase)
- T566-1: mounting flange

## **12 Appendix: T660 and T560/T564 Comparison**

The T660 Digital Delay Generator is Highland Technology's successor to the model T560 and T564 Digital Delay Generators. The T660 strives to maintain complete backwards compatibility with both of these products. Because the T660 adds many new capabilities and functionalities there are some small differences between the T660 and the T560/T564 modules relevant to backwards compatibility.

A comprehensive list of these differences can be found below.

### **12.1 Behavioral Changes for Hardware**

- The T660 takes approximately 10 seconds to power on versus a T560/T564 which takes approximately 3 seconds.
- The T660 can compensate for temperature continuously and dynamically, so unlike the T560, temperature fluctuations do not affect configurations that were applied at different temperatures. This means that frames configurations remain accurate indefinitely after configuration.

### **12.2 Behavioral Changes for ASCII communication protocol**

- T660 is capable of parsing the following characters "+, -, . # /". The T560 and T564 ignored these.
- T660 has the capability to change a channel impedance termination, while the T560/T564 was fixed. Given this capability, the channel configuration status command query adds the termination state to the response as follows:  

```
Ch
A POS ON      Dly  00.000,000,000,000  Wid  00.000,002,000
,000  50R.
```
- All T660 commands do not produce extraneous newline characters. Only multi-line query responses include newline characters and do so in an identical way to the T560/T564. For example, the IDentify query response no longer produces an extra newline character.
- All common line endings are supported and automatically interpreted from all inputs. Thus <cr>, <lf>, and <cr><lf> line endings are all interpreted as a single line delimiter without any glitches.
- A unit refers to itself as a "T660" in every instance the T560/T564 used it model number. Specifically, this difference affects the IDentify and SStatus commands.
- T660 firmware uses a new version numbering scheme, so the IDentify command produces a response in the format T660 DDG Firmware 28E660-1-1.3 instead of T564 DDG Firmware 28E560-E. In this scheme, 1.3 refers to the major and minor firmware version numbers.
- Command parsing has increased in strictness to alert users of easy-to-miss typos that the T560 silently ignored or behaved unexpectedly with. This is only noticeable if programmed not according to the T560 manual.
- Commands are executed much faster than before. Typical command execution speedup is 5-10x.

- The command text buffer is no longer limited to 256 bytes and thus can process far larger command lines. T660 lines should be limited to 16,383 characters.
- Installing settings (via the `INstall` command, for example) results in far less time where triggers are ignored: approximately 1 microsecond for the T660 versus up to 8000 microseconds for a T560.
- All `HElp` commands have been deprecated.

### **12.3 Behavioral Changes for Trains and Frames**

- Frames can operate much faster, at a speed of 1 MHz (compared to the T564's 15 kHz).
- Using the `STatus` command (or the `ASet`, `BSet`, `CSet`, and `DSet` commands) now prints the correct current device state during frames operation. However, since the frames engine may constantly change this data during operation, it is not recommended to rely on this data unless triggers are disabled.
- Some commands listed in the T560/T564 manual that responded in undocumented or undefined ways behave slightly differently. For example, the `RZap` command now guarantees that all frames in the frames memory are initialized to the default (`LOad DEfault`) configuration.